

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 2 – Seminar on Emerging Technologies

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 2 – Seminar on Emerging Technologies
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	y.hema sai lahari	Reg. No.:	192311446
Date:		Duration:	60 minutes

Code of Conduct

I y.hema sai lahari (192311446) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:y.hema sai lahari

Annexure A – Seminar on Emerging Technologies

Instructions to Students:

Students must select the topic **strictly according to the Serial Number (S.No.) order** assigned to them.

S.No.	Seminar Topic
1	Generative AI and Large Language Models (LLMs)
2	AI Agents and Autonomous AI Systems

3	Multimodal Artificial Intelligence
4	Explainable and Trustworthy AI
5	Edge AI and Intelligent Edge Computing
6	Federated Learning and Privacy-Preserving AI
7	Deep Learning for Computer Vision
8	AI-Powered Medical Image Analysis
9	Natural Language Processing with Transformer Models
10	Retrieval-Augmented Generation (RAG)
11	AI in Cybersecurity and Threat Detection
12	Zero Trust Architecture and Modern Cybersecurity
13	Post-Quantum Cryptography
14	Quantum Computing and Quantum Algorithms
15	Quantum Machine Learning
16	Cloud-Native Computing and Microservices
17	Serverless Computing and Function-as-a-Service
18	Edge Computing and Fog Computing
19	Cloud Security and Confidential Computing
20	Digital Twins and Intelligent Simulation
21	Internet of Things (IoT) and Smart Systems
22	Industrial Internet of Things (IIoT)
23	5G/6G Networks and Next-Generation Communication
24	Blockchain Technology and Decentralized Applications
25	Web3 and Decentralized Internet
26	Smart Contracts and Blockchain Applications
27	Augmented Reality (AR) and Virtual Reality (VR)
28	Extended Reality (XR) and Immersive Technologies
29	Robotics and Autonomous Robots
30	Human–Robot Interaction and Collaborative Robots

31	Autonomous Vehicles and Intelligent Transportation Systems
32	Smart Cities and AI-Driven Urban Technologies
33	Metaverse Technologies and Virtual Worlds
34	Brain–Computer Interfaces (BCI)
35	Wearable Computing and Smart Wearable Devices
36	Green Computing and Sustainable IT
37	AI for Software Engineering and Code Generation
38	DevSecOps and Intelligent CI/CD Pipelines
39	Digital Forensics and AI-Based Cybercrime Investigation
40	Privacy-Enhancing Technologies and Data Protection

Seminar Presentation Structure:

1. **Introduction to the Emerging Technology**
2. **Need and Motivation**
3. **Working Principle / Architecture**
4. **Key Components and Technologies**
5. **Applications / Real-World Use Cases**
6. **Advantages and Limitations**
7. **Recent Developments and Trends**
8. **Challenges and Future Scope**
9. **Case Study / Example**
10. **Conclusion and References**

SEMINAR PRESENTATION

Privacy-Enhancing Technologies and Data Protection

1. Introduction to the Emerging Technology

Privacy-Enhancing Technologies (PETs) are technologies and techniques designed to protect personal and sensitive information while allowing data to be processed and used for

legitimate purposes. PETs reduce unnecessary exposure of personal data and help organizations follow privacy and data-protection requirements.

Traditional security mainly focuses on preventing unauthorized access, while PETs additionally focus on **minimizing the amount of personal information that is revealed or processed**.

Important Privacy-Enhancing Technologies include:

- Differential Privacy
- Homomorphic Encryption
- Secure Multi-Party Computation
- Federated Learning
- Zero-Knowledge Proofs
- Data Anonymization and Pseudonymization

PETs are becoming increasingly important with the growth of cloud computing, artificial intelligence, IoT, digital payments, and large-scale data analytics.

2. Need and Motivation

Organizations collect large amounts of personal data such as names, locations, financial information, health information, and online activity. Improper handling of this information can lead to privacy violations, identity theft, data breaches, and misuse of personal information.

The major motivations for using PETs are:

1. To protect sensitive personal information.
 2. To reduce the risk of data breaches.
 3. To minimize unnecessary collection and exposure of data.
 4. To enable secure data analysis without revealing sensitive information.
 5. To improve user trust in digital services.
 6. To support privacy and data-protection regulations.
 7. To enable organizations to use data while maintaining individual privacy.
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3. Working Principle / Architecture

Privacy-enhancing systems generally follow the principle of **collecting, processing, and sharing only the information that is necessary**.

Basic Architecture

User Data → Privacy Protection Mechanism → Secure Processing → Result → Authorized User

Working Process

1. The user provides data to the system.
2. A privacy mechanism protects or transforms the data.
3. The protected data is processed or analyzed.
4. Only the required information or result is provided.
5. Sensitive raw data is minimized or kept hidden from unauthorized parties.

Different PETs use different mechanisms:

- **Differential Privacy:** Adds controlled statistical noise to protect individuals.
 - **Homomorphic Encryption:** Allows computation directly on encrypted data.
 - **Federated Learning:** Trains models using data stored on individual devices without directly sharing raw data.
 - **Secure Multi-Party Computation:** Allows multiple parties to jointly compute a result without revealing their private inputs.
 - **Zero-Knowledge Proofs:** Allows a person to prove that something is true without revealing the underlying secret.
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4. Key Components and Technologies

4.1 Differential Privacy

Differential Privacy protects individuals by adding carefully controlled noise to query results or datasets. It reduces the possibility of identifying a specific person's information.

4.2 Homomorphic Encryption

Homomorphic Encryption allows computations to be performed on encrypted data. The data does not need to be decrypted before processing.

4.3 Federated Learning

Federated Learning trains machine-learning models across multiple devices or organizations while keeping the original data locally.

4.4 Secure Multi-Party Computation

Secure Multi-Party Computation allows several parties to calculate a common result without exposing their individual private data.

4.5 Zero-Knowledge Proofs

Zero-Knowledge Proofs allow one party to prove the validity of information without revealing the actual information.

4.6 Anonymization and Pseudonymization

Anonymization removes identifying information from data, while pseudonymization replaces direct identifiers with artificial identifiers.

5. Applications / Real-World Use Cases

Privacy-Enhancing Technologies can be applied in many areas.

Healthcare

PETs can enable researchers and hospitals to analyze medical information while reducing exposure of individual patient data.

Banking and Finance

Financial institutions can use privacy-preserving techniques for fraud detection, risk analysis, and secure data sharing.

Artificial Intelligence

Federated Learning and Differential Privacy can help develop AI models while reducing the need to centralize sensitive user data.

Smart Cities

PETs can help analyze traffic, transportation, and public-service data while protecting citizens' personal information.

Digital Identity

Zero-Knowledge Proofs can allow users to prove specific attributes without revealing unnecessary personal information.

Advertising

Privacy-preserving analytics can help measure advertising performance while reducing the exposure of individual user behavior.

Cloud Computing

Homomorphic Encryption and other PETs can help organizations process sensitive information in cloud environments without directly exposing the underlying data.

6. Advantages and Limitations

Advantages

1. Improves protection of personal information.
2. Reduces privacy risks and unnecessary data exposure.
3. Supports secure data sharing and analysis.
4. Increases user trust in digital services.
5. Helps organizations meet privacy requirements.
6. Enables privacy-preserving artificial intelligence and analytics.
7. Reduces the impact of certain data breaches.

Limitations

1. Some PETs require significant computational resources.
2. Privacy mechanisms can reduce data accuracy or utility.
3. Implementation can be technically complex.
4. Advanced cryptographic techniques may increase processing time.
5. Organizations may require specialized skills and infrastructure.
6. Balancing privacy, performance, cost, and usability can be difficult.

7. Recent Developments and Trends

Privacy-enhancing technologies are developing rapidly because of increasing concerns about personal-data protection and AI.

Important trends include:

Privacy-Preserving AI

Researchers are combining PETs with AI to develop machine-learning systems that can process sensitive information while reducing data exposure.

Federated Learning

Federated Learning is increasingly being explored for healthcare, finance, mobile applications, and IoT because data can remain closer to its source.

Advanced Encryption

Research is improving the efficiency and practicality of homomorphic encryption and other cryptographic techniques.

Zero-Knowledge Technologies

Zero-knowledge proofs are becoming increasingly important for privacy-preserving digital identity, authentication, and blockchain applications.

Privacy by Design

Organizations are increasingly incorporating privacy protection into systems from the beginning rather than adding privacy controls after development.

Regulatory Technology

PETs are increasingly being considered as tools that can help organizations implement data-minimization and privacy requirements.

8. Challenges and Future Scope

Challenges

1. High computational and storage requirements.
2. Difficulty in balancing privacy and data utility.
3. Complex implementation and maintenance.
4. Lack of common standards across different PET solutions.
5. Limited awareness and expertise among organizations.
6. Scalability issues for large datasets and large numbers of users.
7. Integration with existing IT systems can be difficult.

Future Scope

1. Development of faster and more efficient privacy-preserving algorithms.
 2. Greater integration of PETs with Artificial Intelligence.
 3. Improved privacy-preserving cloud computing.
 4. Wider adoption of federated learning.
 5. More practical homomorphic encryption solutions.
 6. Privacy-preserving digital identity systems.
 7. Automated privacy management and monitoring.
 8. Development of standardized frameworks for PET implementation.
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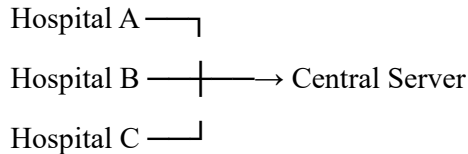
9. Case Study / Example

Case Study: Federated Learning for Healthcare

Consider several hospitals that want to develop an AI model for disease analysis.

Traditional Approach

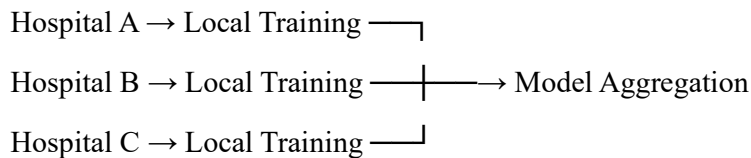
Each hospital sends patient data to a central server.



This requires transferring sensitive patient information and increases privacy risks.

Privacy-Preserving Approach

Using Federated Learning, each hospital keeps its patient data locally.



Working

1. A common AI model is distributed to participating hospitals.
2. Each hospital trains the model using its local data.
3. The hospitals send model updates rather than raw patient records.
4. The updates are aggregated to improve the shared model.
5. Patient data remains within the respective hospitals.

Benefits

- Reduces direct sharing of raw patient data.
- Supports collaborative machine learning.
- Improves privacy protection.
- Allows organizations to benefit from combined knowledge.

Limitation

Federated Learning can still face risks such as model-update leakage, communication overhead, heterogeneous data, and attacks against the training process. Additional privacy and security mechanisms may therefore be required.

10. Conclusion and References

Conclusion

Privacy-Enhancing Technologies provide an important approach to protecting personal information while still enabling useful data processing and analysis. Technologies such as Differential Privacy, Homomorphic Encryption, Federated Learning, Secure Multi-Party Computation, and Zero-Knowledge Proofs provide different methods for reducing data exposure.

As organizations increasingly depend on artificial intelligence, cloud computing, IoT, and large-scale data analytics, privacy protection has become an essential requirement. PETs can help create systems that balance **data utility, security, and privacy**.

In the future, improved algorithms, better standards, and integration with AI and cloud technologies are expected to make PETs more practical and widely adopted.

References

- [1] C. Dwork and A. Roth, “The Algorithmic Foundations of Differential Privacy,” *Foundations and Trends in Theoretical Computer Science*, vol. 9, nos. 3–4, pp. 211–407, 2014.
- [2] C. Gentry, “Fully Homomorphic Encryption Using Ideal Lattices,” in *Proc. 41st Annual ACM Symposium on Theory of Computing (STOC)*, 2009, pp. 169–178.
- [3] B. McMahan et al., “Communication-Efficient Learning of Deep Networks from Decentralized Data,” in *Proc. 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017.
- [4] A. C. Yao, “Protocols for Secure Computations,” in *Proc. 23rd Annual Symposium on Foundations of Computer Science*, 1982, pp. 160–164.
- [5] J. Camenisch and M. Stadler, “Proof Systems for General Statements about Discrete Logarithms,” Technical Report, ETH Zürich, 1997.
- [6] NIST, “Privacy-Enhancing Cryptography,” National Institute of Standards and Technology.
- [7] European Union Agency for Cybersecurity (ENISA), “Data Protection Engineering,” guidance and technical resources on privacy-enhancing technologies.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
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Technical Content & Subject Knowledge (25%)	Comprehensive and accurate content; demonstrates excellent understanding of the emerging technology	Good technical understanding with minor gaps	Basic understanding with limited technical depth	Incomplete content or major technical errors
Organization & Presentation Structure (15%)	Excellent logical flow with clear introduction, content, conclusion, and effective transitions	Well-structured with minor organizational issues	Adequately structured but some sections lack clarity	Poorly structured and difficult to follow
Examples, Diagrams & Case Study (20%)	Excellent use of relevant diagrams, examples, applications, and real-world case studies	Good use of relevant examples and diagrams	Limited examples/diagrams with basic explanation	Inappropriate or insufficient examples/diagrams
Communication & Presentation Skills (20%)	Very clear, confident, engaging delivery with excellent eye contact, voice modulation, and time management	Clear and confident presentation with minor issues	Understandable but lacks confidence or engagement	Unclear delivery, excessive reading, or poor time management
Question Handling & Overall Performance (20%)	Answers questions accurately and confidently; demonstrates	Answers most questions correctly	Answers basic questions with some difficulty	Unable to answer questions or demonstrate adequate understanding

	strong overall knowledge			
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Criteria	Mark
Technical Content & Subject Knowledge (25%)	/5
Organization & Presentation Structure (15%)	/5
Examples, Diagrams & Case Study (20%)	/5
Communication & Presentation Skills (20%)	/5
Question Handling & Overall Performance (20%)	/5
TOTAL	/25